

Problem Statement

Interview Trainer Agent – AI-Powered Role-Based Interview Preparation Coach

Domain	HR Tech / EdTech / AI-Based Career Preparation
Project Name	Interview Trainer Agent
Core Technologies	IBM watsonx.ai, IBM Granite Models, IBM Cloud Lite, React, Node.js, Express, PDF Parsing, OCR

The Challenge

Job seekers often prepare for interviews using generic question banks, copied answers, or fixed online practice sets. These resources do not consider the candidate's actual resume, projects, technical skills, experience level, or target job role. As a result, candidates may practice questions that are irrelevant, over-polished, or disconnected from what they can genuinely explain in an interview.

This problem is especially common for students, freshers, and early-career applicants who may have strong project work but limited interview experience. They need a simple, affordable, and personalized preparation assistant that can read their resume, understand their profile, generate relevant questions, provide natural model answers, and suggest practical resume improvements.

The Objective

The objective of this project is to build an AI-powered Interview Trainer Agent that works as a personalized virtual interview coach. The system should accept a resume in PDF or image format, extract the candidate's information, understand the target role, and generate concise interview preparation content in a clean question-and-answer format.

- Extract readable text from PDF resumes and image-based resumes using PDF parsing and OCR.
- Identify key candidate details such as role, experience level, education, skills, projects, and certifications.
- Generate 5 to 10 role-specific interview questions, with each question written in one line.
- Provide short model answers of 2 to 3 lines that sound natural, student-friendly, and resume-aligned.
- Cover technical, project-based, behavioral, and HR-style interview preparation areas.
- Give practical suggestions to improve resume keywords, project clarity, and role alignment.
- Use IBM watsonx.ai and IBM Granite models to generate contextual, structured, and real-time responses.

Expected Solution Outcome

The Interview Trainer Agent aims to deliver an accessible and intelligent interview preparation platform that helps candidates practice smarter. Instead of acting like a static question bank, it behaves like a digital coach that adapts questions and answers according to the candidate's resume and target role.

The final output should be easy to read and directly useful for preparation: a detected profile type, resume score, interview questions, short answers, and focused suggestions. This helps users improve confidence,

understand role expectations, and prepare responses that are realistic and aligned with their own background.

Key Functional Modules

Module	Purpose
Resume Parser	Extracts text from uploaded PDF resumes and converts image resumes into text using OCR.
Profile Analyzer	Identifies candidate details such as role, experience level, education, skills, projects, and certifications.
Interview Question Generator	Generates 5 to 10 role-specific questions covering technical, project, behavioral, and HR areas.
Answer Coach	Creates short, natural, first-person model answers that are based only on the candidate's provided profile.
Resume Feedback Engine	Provides a resume score and practical suggestions related to keyword match, project clarity, communication, role fit, and readiness.

System Workflow

1	User uploads a resume or manually enters profile and target role details.
2	Backend extracts and cleans resume text from PDF or image input.
3	Prompt builder structures the candidate profile and sends it to IBM Granite through watsonx.ai.
4	Granite generates structured JSON containing profile type, score, questions, answers, and suggestions.
5	React frontend displays the interview set in a clean practice-friendly interface.

Disclaimer

Interview Trainer Agent provides general interview preparation support only. It does not guarantee job selection, replace professional career counseling, or verify the accuracy of all hiring expectations. Users should customize the generated answers with their real experiences before using them in interviews.

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